HW2

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Market Data Analysis

- Dec. 2027 futures imply a rate of 3.415%
- This suggests only ~21 bps of easing, less than one full rate cut
- FedWatch is concentrated around the 3.00%–3.50% target range
- Takeaway: markets expect a modest easing cycle

	CME FEDWATCH TOOL - CONDITIONAL MEETING PROBABILITIES							
MEETING DATE	225-250	250-275	275-300	300-325	325-350	350-375	375-400	400-425
2026/4/29	0.0%	0.0%	0.0%	0.0%	0.0%	100.0%	0.0%	0.0%
2026/6/17	0.0%	0.0%	0.0%	0.0%	3.2%	96.8%	0.0%	0.0%
2026/7/29	0.0%	0.0%	0.0%	0.1%	7.2%	92.6%	0.0%	0.0%
2026/9/16	0.0%	0.0%	0.0%	0.4%	10.8%	88.7%	0.0%	0.0%
2026/10/28	0.0%	0.0%	0.0%	0.4%	10.6%	87.0%	2.0%	0.0%
2026/12/9	0.0%	0.0%	0.0%	0.4%	10.6%	87.0%	2.0%	0.0%
2027/1/27	0.0%	0.0%	0.0%	0.4%	9.8%	80.8%	8.8%	0.2%
2027/3/17	0.0%	0.0%	0.0%	0.4%	9.0%	74.8%	14.9%	0.9%
2027/4/28	0.0%	0.0%	0.0%	0.3%	8.4%	69.9%	19.4%	1.9%
2027/6/9	0.0%	0.0%	0.0%	0.9%	12.4%	66.6%	18.3%	1.8%
2027/7/28	0.0%	0.0%	0.1%	1.4%	14.8%	64.4%	17.5%	1.7%
2027/9/15	0.0%	0.0%	0.3%	4.0%	24.4%	55.4%	14.5%	1.4%
2027/10/27	0.0%	0.1%	1.4%	9.8%	33.3%	43.6%	10.7%	1.0%
2027/12/8	0.2%	1.8%	11.0%	33.8%	41.9%	10.2%	1.0%	0.0%

Table: FedWatch-Target Rate Probability

Historical Data Analysis

Cutting periods

Bond perform best, returns 6.37% with 62% chance of positive monthly return
Others all negatively returned

Non-cutting periods

Equity perform the best, earning 15.46% with 65% of positive monthly return

	Cutting Ann. Ret %	Cutting Ann. Vol %	Cutting % months > 0	Non-Cutting Ann. Ret %	Non-Cutting Ann. Vol %	Non-Cutting % months > 0
US Equity	-6.62	20.19	31.6	15.46	13.36	65.2
US Bond	6.37	8.40	62.1	4.65	7.10	57.2
Commodities	-14.72	26.40	17.9	5.36	16.44	28.9
US HY Credit	-3.31	18.99	14.7	6.85	7.75	36.3

Table: Asset Performance in Fed Cutting vs Non-Cutting since 1990

- Conclusion: Fed cuts more associated with economic stress than broad risk-asset rallies.

Does this Make Sense for Now? Partly.

- **Bond:** Supported by easing, but limited upside with only ~0.84 cuts priced
- **Equity:** Conditional. Rate cuts often hurt equities when recession-driven, but can still support gains in a soft landing.
- **Commodity/HY Credit:** Historically weak in cutting periods; performance depends on sustained growth and risk appetite than rate cuts

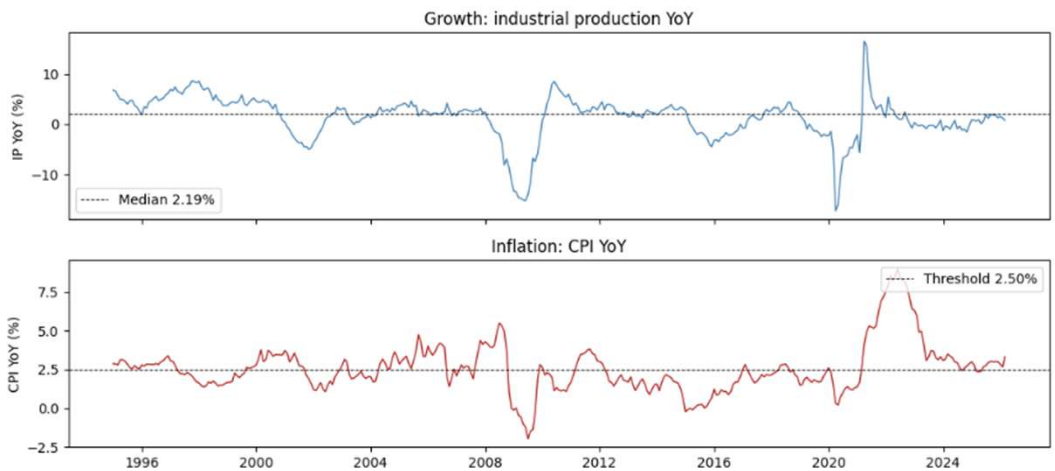
Conclusion: Long duration is the cleanest Fed easing expression, while other risk assets require a constructive growth outlook and should not rely on cuts alone

Macro Regimes

Macro Regimes: Growth × Inflation Framework

Growth (Industrial Production yoy)	Inflation (CPI yoy)
Threshold = **2.19%**	Threshold = **2.50%**

Build Regime



Plot: Growth and Inflation Thresholds for Macro Regime

	Inflation Falling (Dn)	Inflation Rising (Up)
Growth Rising (Up)	**Goldilocks** *High Growth / Low Inflation* **Historical:** Late 90s, mid-2010s **Market:** Equities & Credit outperform.	**Reflation / Overheating** *High Growth / High Inflation* **Historical:** 2021 Post-COVID **Market:** Commodities & Value stocks.
Growth Falling (Dn)	**Disinflationary Slowdown** *Low Growth / Low Inflation* **Historical:** 2008 GFC, 2015 Slump **Market:** Bonds & Defensive assets.	**Stagflation** *Low Growth / High Inflation* **Historical:** 1970s, parts of 2022 **Market:** Cash & Inflation-linked bonds.

- Periods shown require at least six consecutive months in the same regime to reduce short-term noise.

Macro Regimes

Asset Performance in Regime

Equity: overall strong except Stagflation

Bond: strongest in growth-up

Commodity: Strong in Reflation, weak elsewhere

HY Credit: strong in Goldilocks/Deflation

- **Growth↑ ---** equity/bond
- **Inflation↓** --- equity/HY credit
- **Growth↑** & **Inflation↑** --- commodity
- Stagflation is worst

	Ann Mean %				Ann Vol %				Sharpe			
Asset	Commodities	US Bond	US Equity	US HY Credit	Commodities	US Bond	US Equity	US HY Credit	Commodities	US Bond	US Equity	US HY Credit
regime												
GrowDn/InflDn	1.82	3.48	13.83	10.41	19.09	8.34	15.91	12.18	0.10	0.42	0.87	0.85
GrowDn/InflUp	-6.38	1.40	0.07	-0.97	19.46	7.63	17.14	11.67	-0.33	0.18	0.00	-0.08
GrowUp/InflDn	-2.86	5.91	16.18	5.31	16.63	6.87	14.99	7.05	-0.17	0.86	1.08	0.75
GrowUp/InflUp	18.60	6.51	12.44	-0.17	17.78	6.93	12.64	8.53	1.05	0.94	0.98	-0.02

Table: Macro Asset Performance in Regimes since 1995

Same Risk Metrics for All Regimes? No.

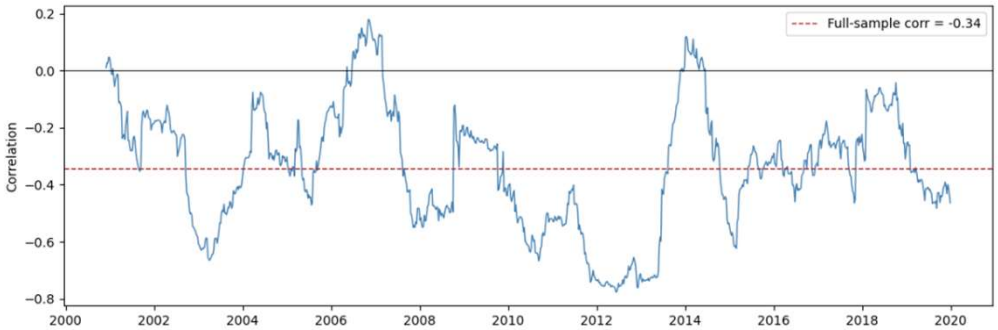
- Goldilocks / Reflation: **Volatility/Sharpe** capture most key risk-return dynamics
- Stagflation: emphasize stock-bond **correlation** and inflation **beta**
- Deflationary Slowdown: emphasize **max drawdown, CVaR**
- For all regimes, could also evaluate **macro vol-of-vol, VIX, and MOVE**

Mini-project question

Equity-Bond Correlation

- Most of time- strong negative correlation
- But when inflation dominates- weakens or even positive
- Stock–bond diversification is generally powerful but not always

2000-2019 Full-sample	1 Year Rolling
ρ = -0.34 Equity/Bond move in opposite directions	Mean = **-0.33** Ranging (**−0.78, 0.18**) **93%** of time negative

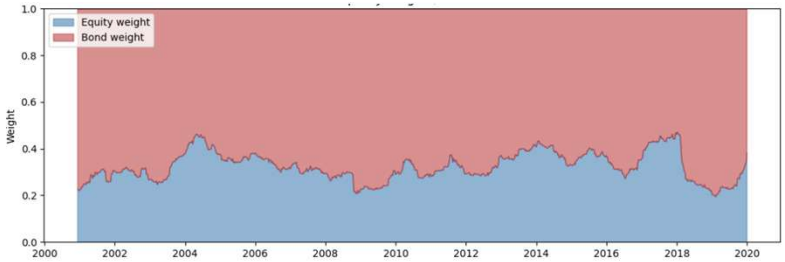


Plot: Rolling 1-year equity-bond return weekly correlation

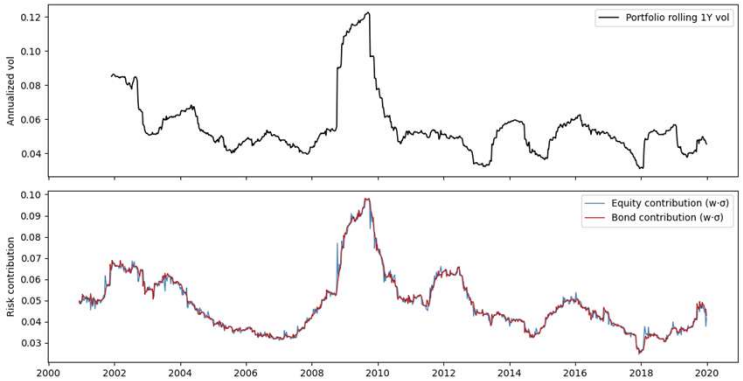
Mini-project question

Risk Parity Performance

	Equity	Bond	Reasoning
Allocation	33%	67%	Bond weight more due to less volatility
Risk contribution to portfolio vol 5.5%	4.85%	4.86%	Equally distributed and diversified risk
Metrics	• Return = 6% • Sharpe = 1.0		1. Negative correlation → diversification 2. Similar Sharpe across assets 3. Risk-based allocation avoids equity dominance



Plot: Risk-Parity Weights, 2000-2019



Plot: Risk-Parity Fund Risk Over Time

Mini-project question

2021 – Present

When Risk Parity Fails

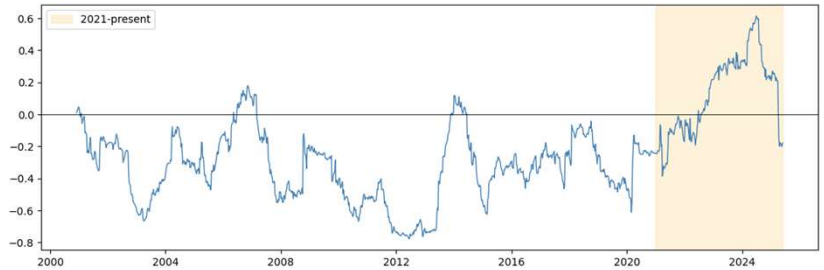
1. Risk Parity Performance

- Max drawdown **−16.8%** in 2022
- Cumulative return only **4.2%** after recovery

2. Why it failed?

- Equity-Bond Correlation: negative→ **positive**
- Risk diversification → risk **concentration**

Plot: Rolling 1-year equity-bond return weekly correlation



3. Implications

- Do not always assume **negative ρ**, use faster risk controls when it flips positive
- Use regime-aware risk allocation (inflation vs. growth)
- Add inflation hedges (commodity, TIPS, gold)